

ESTIMATION PRO — FIXED RATE BAND COST ESTIMATE

Project: Quick Analysis

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Model: STRUCTMIND CORE · PRO

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CALCULATION MANIFEST — SINGLE SOURCE OF TRUTH

--- All locked input and output values for this estimate. Sections 1 and 7 copy from here.

Parameter	Value
AI-Extracted Fabricated Tonnage	12.33 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Material Factor	x1.00
Surface Factor	x1.00
Assembly Factor	x1.10
Composite Adjustment Factor	x1.10
Fixed Rate — Low (USD per ton)	\$3,500
Fixed Rate — Mid (USD per ton)	\$4,000
Fixed Rate — High (USD per ton)	\$4,500
Adjusted Rate — Low	\$3,850
Adjusted Rate — Mid	\$4,400

Parameter	Value
Adjusted Rate — High	\$4,950
Country	USA
Tax Rate Applied	0%
LOCK COST — LOW	\$47,500
LOCK COST — MID	\$54,300
LOCK COST — HIGH	\$61,000

1. EXECUTIVE SUMMARY

--- Project cost summary at fixed rate band.

Metric	Value
Project Name or ID	HOME 2 - CAROWINDS CABELAS DR.
Country	USA
Total Fabricated Tonnage	12.33 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Composite Adjustment Factor	x1.10
Fixed Rate Band Applied	\$3,500 — \$4,500 per ton
Tax Rate	0%
Estimated Cost — Low (inc. tax)	\$47,500
Estimated Cost — Mid (inc. tax)	\$54,300

Metric	Value
Estimated Cost — High (inc. tax)	\$61,000
Cost per Fabricated Ton — Mid	\$4,400 per ton

Top Three Commercial Risks: 1. MISSING DATA — Column lengths, lintel quantities/lengths, and bearing plate quantities are absent from the drawings and have been assumed for this estimate. This represents a potential tonnage variance of +/- 2.5 tons. 2. PROFILE DISCREPANCIES — Conflicts exist between schedules and plans (e.g., C86 vs C8). HSS6x6x5/8 is a non-standard profile requiring manual weight verification. Resolution will require RFIs and may alter tonnage. 3. SCOPE GAPS — The full scope for loose brick lintels and galvanized bearing plates is undefined. Allowances are included, but actual requirements could significantly impact cost for materials and galvanizing.

2. BASIS OF ESTIMATE

--- Drawings reviewed: S0.01, S0.02, S1.01, S1.02, S1.04, S2.01, S2.02, S2.03, S2.04, S3.01, S3.02, S3.03, S3.04, S3.05, S3.06.

Tonnage extraction method: (c) Estimated from framing plans where BOM is absent or incomplete. Member lengths were derived from grid dimensions on drawing S1.02. Column heights and lintel quantities/lengths were assumed due to missing information.

Composite factor selection:

- Material (Carbon Steel): Explicitly stated in General Notes on sheet S0.01, Section 051200, item 1.04.
- Surface Treatment (Shop primer only): Stated as the primary coating in General Notes on sheet S0.01, Section 051200, item 2.05.A. A note on S0.02 requires some bearing plates to be galvanized, which is flagged as a conflict.
- Assembly Complexity (Mixed bolt and weld): Inferred from connection details on sheet S0.02 and S3.06, which show a combination of bolted connections, welded plates, and welded studs.

Rate basis: Fixed rate band of \$3,500 (low) to \$4,500 (high) per fabricated ton applied. Midpoint rate of \$4,000 per fabricated ton used for the headline cost. These rates represent current USA structural steel shop fabrication pricing for the member types and complexity class identified in this package.

Tax applied: 0% for USA.

Global assumptions: All structural steel grades conform to notes on S0.01 (A992 for W-shapes, A500 Gr. C for HSS, A36 for plates/angles).

3. MEMBER-BY-MEMBER TONNAGE TAKE-OFF

--- Table 3A — Primary Structural Members Columns, primary beams, moment frames, transfer members, and lateral bracing.

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (m)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
1	Column	C5	HSS6x6 x3/8	2	14'-0"	4,267	33.1	282.4	622.6	A500 Gr. C	S0.02, S1.01	Assumed Length
2	Column	C6 /C8	HSS6x6 x5/8	6	14'-0"	4,267	NOT IN TABLE	1,285.4	2,833.8	A500 Gr. C	S0.02, S1.01	Assumed Length
3	Column	CSA	HSS6x6 x3/8	4	14'-0"	4,267	33.1	564.9	1,245.4	A500 Gr. C	S0.02, S1.01	Assumed Length
4	Column	Trellis	HSS8x6 x3/8	4	8'-0"	2,438	NOT IN TABLE	386.2	851.4	A500 Gr. C	S3.06	Assumed Length
5	Beam	—	W14x22	32	12'-10"	3,912	32.7	4,096.3	9,030.8	A992	S1.02	From Plan
6	Beam	—	W16x36	1	25'-3"	7,696	53.6	412.5	909.4	A992	S1.02	From Plan
7	Beam	—	HSS18x6x3/8	7	Various	—	NOT IN TABLE	1,693.6	3,733.7	A500 Gr. C	S1.02	From Plan

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (m)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
8	Beam	Trellis	HSS8x6x3/8	2	13'-0"	3,962	NO T IN TA BLE	313.8	691.8	A500 Gr. C	S3.06	From Plan
9	Lintel	L1	W8x24	2	20'-0"	6,096	35.7	435.2	959.5	A992	S0.02	Assumed Qty/Len
10	Lintel	L2	W16x50	2	20'-0"	6,096	74.4	907.3	2,000.3	A992	S0.02	Assumed Qty/Len

Table 3B — Secondary and Miscellaneous Members Secondary beams, purlins, girts, joists, bracing, stairs, handrails, and platforms.

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (m)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
11	Beam	—	HSS9x5x3/8	1	12'-10"	3,912	NO T IN TA BLE	136.1	300.1	A500 Gr. C	S1.02	From Plan
12	Beam	—	HSS6x4x1/4	1	12'-10"	3,912	18.0	70.4	155.2	A500 Gr. C	S1.02	From Plan

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
13	Lintel	—	L6x4x3/8	20	6'-0"	1,829	14.7	537.7	1,185.4	A36	S0.02	Assumed Qty/Length

Table 3C — Connection Material and Plates Shear plates, base plates, stiffeners, gusset plates, and embedded plates.

No.	Description	Thickness	Width	Length	Qty	Weight (kg)	Weight (lb)	Source Sheet	Extraction Method
14	Base Plate for C5	1"	12"	12"	2	37.0	81.6	S0.02	From Schedule
15	Base Plate for C6/C8	1"	14"	14"	6	151.2	333.3	S0.02	From Schedule
16	Base Plate for CSA	3/4"	12"	12"	4	55.6	122.6	S0.02	From Schedule
17	Base Plate for Trellis	1"	16"	16"	4	131.6	290.1	S3.06	From Detail
18	Bottom Plate for L1	1/4"	7"	40'-0"	1	108.0	238.1	S0.02	Assumed
19	Bottom Plate for L2	1/4"	10"	40'-0"	1	154.3	340.2	S0.02	Assumed

No.	Description	Thickness	Width	Length	Qty	Weight (kg)	Weight (lb)	Source Sheet	Extraction Method
20	Bearing Plate Allowance	3/4"	8"	12"	20	184.0	405.7	S0.02	Assumed
21	Cap Plate for WP	1/2"	10"	10"	4	25.6	56.4	S0.02	Assumed

Table 3D — Tonnage Subtotals Consolidated weight by category feeding into the cost model.

Category	Count	Total Weight (kg)	Total Weight (lb)	Est. Tons
W-Shapes — Columns and Beams	38	5,851.3	12,899.9	5.85
HSS and Tube	22	4,732.8	10,434.4	4.73
Pipe	0	0.0	0.0	0.00
Angles and Channels	20	537.7	1,185.4	0.54
Plates (all types)	41	847.3	1,867.9	0.85
Miscellaneous and Embeds	0	0.0	0.0	0.00
Bolts and Accessories (3.00% addition)	—	359.1	791.7	0.36
TOTAL	121	12,328.2	27,179.3	12.33

4. COST BUILD-UP BY MEMBER CATEGORY

--- Per-category cost using the three-rate band. Adjusted rates = base rate x composite factor.

Category	Est. Tons	Adj. Rate Low (USD/ton)	Adj. Rate Mid (USD/ton)	Adj. Rate High (USD/ton)	Cost Low (USD)	Cost Mid (USD)	Cost High (USD)	Pct of Total
W-Shapes	5.85	\$3,850	\$4,400	\$4,950	\$22,523	\$25,740	\$28,958	47.5%
HSS and Tube	4.73	\$3,850	\$4,400	\$4,950	\$18,211	\$20,812	\$23,414	38.4%
Pipe	0.00	\$3,850	\$4,400	\$4,950	\$0	\$0	\$0	0.0%
Angles and Channels	0.54	\$3,850	\$4,400	\$4,950	\$2,079	\$2,376	\$2,673	4.4%
Plates	0.85	\$3,850	\$4,400	\$4,950	\$3,273	\$3,740	\$4,208	6.9%
Miscellaneous and Embeds	0.00	\$3,850	\$4,400	\$4,950	\$0	\$0	\$0	0.0%
Bolts and Accessories	0.36	\$3,850	\$4,400	\$4,950	\$1,386	\$1,584	\$1,782	2.9%
TOTAL	12.33	\$3,850	\$4,400	\$4,950	\$47,471	\$54,252	\$61,034	100%

5. PROCESS COST BREAKDOWN — MIDPOINT SCENARIO

--- Distributing \$54,300 across industry-standard fabrication process activities. Percentages are fixed fabrication industry standards applied at midpoint.

Fabrication cost allocated across process activities at the midpoint scenario.

Process Activity	Industry Pct	Amount (USD)	Notes
Material — mill sections and plate	35%	\$18,900	Structural steel mill cost
Shop labour — cut, fit, and weld	30%	\$16,300	Includes AWS D1.1 weld operations

Process Activity	Industry Pct	Amount (USD)	Notes
Coatings and surface preparation	12%	\$6,500	SSPC class as specified
Consumables — weld wire, bolts, and gas	6%	\$3,300	Per-ton allowance
Equipment — plasma, burn table, drill	7%	\$3,800	CNC and NC machine allocation
Overhead and QA/QC	6%	\$3,300	Inspection, NDT, project management
Shop margin	4%	\$2,200	Fabricator margin
TOTAL	100%	\$54,300	Midpoint scenario

6. DRAWING COMPLETENESS AND COST EXPOSURE ASSESSMENT

--- Areas of drawing incompleteness that create commercial exposure for this estimate.

Area of Assessment	Status	Commercial Exposure if Unresolved
All quantities confirmed from BOM or member schedule	NOT CONFIRMED	CRITICAL: Tonnage may vary by +/- 20% due to assumed quantities and lengths.
Material grades explicit on all members	CONFIRMED	Low: Grades are clear in General Notes on S0.01.
Surface treatment confirmed, not assumed	PARTIAL	Medium: Note for galvanized plates (S0.02) conflicts with general primer note (S0.01).
Assembly complexity confirmed from drawings	CONFIRMED	Low: Details on S0.02 and S3.06 are sufficient to determine complexity.

Area of Assessment	Status	Commercial Exposure if Unresolved
No conflicting BOM vs general notes grade callouts	CONFIRMED	Low: No conflicts found.
Drawings are IFC or internally coordinated	PARTIAL	CRITICAL: Significant conflicts between plans and schedules create risk.
Galvanizing, AESS, or fire rating fully resolved	NOT CONFIRMED	Medium: Scope of galvanizing is unclear.

Cost Waterfall — Midpoint Scenario Step-by-step cost build-up at the midpoint rate.

Step	Amount (USD)
Base cost — 12.33t x \$4,400/ton	\$54,252
Tax at 0%	+\$0
Grand Total — Low (at \$3,500/ton base x composite x tax)	\$47,500
Grand Total — Mid — Headline Cost	\$54,300
Grand Total — High (at \$4,500/ton base x composite x tax)	\$61,000

7. COST CONVERSION SUMMARY

--- Final locked cost summary for this estimate.

Field	Value
Fabricated Tonnage	12.33 t
Fixed Rate Band Applied	\$3,500 — \$4,500 per ton
Composite Adjustment Factor	x1.10

Field	Value
Adjusted Rate — Low	\$3,850 per ton
Adjusted Rate — Mid	\$4,400 per ton
Adjusted Rate — High	\$4,950 per ton
Country and Tax	USA at 0%
Grand Total — Low (inc. tax)	\$47,500
Grand Total — Mid — Headline (inc. tax)	\$54,300
Grand Total — High (inc. tax)	\$61,000
Cost per Fabricated Ton — Mid	\$4,400 per ton

8. ASSUMPTIONS AND EXCLUSIONS

--- Key Assumptions (minimum five specific items — cite sheet references):

- Column heights are assumed to be 14'-0" (4,267 mm) for main columns and 8'-0" (2,438 mm) for trellis columns, as no elevations or lengths are provided in schedules on S0.02.
- Rate band of \$3,500 to \$4,500 per fabricated ton represents current USA market pricing for the member types, surface treatment, and assembly complexity identified in this estimate.
- Quantities and lengths for all lintels (W-shape and angle) are assumed based on typical building requirements, as they are not quantified or located on framing plans (see S0.02 and S1.02).
- Unit weights for profiles not found in the standard reference table (e.g., HSS6x6x5/8) have been sourced from external AISC data and require verification before procurement.
- All conflicts between plans and schedules (e.g., column marks C8 vs. C86 on S1.01 and S0.02) have been resolved by prioritizing the schedule, pending formal RFI clarification.
- The primary coating is shop primer per S0.01; the cost impact of galvanized bearing plates noted on S0.02 is assumed to be minimal and contained within the estimate band.

Exclusions — Items NOT included in this estimate:

- Site erection labour, cranes, manlifts, rigging, and flagging.
- Professional engineer stamping or delegated connection design.
- Loose anchor bolts cast-in-place by the general contractor.

- Site touch-up painting after final erection.
- Tax or duties beyond 0% applied in this estimate.
- 3D MEP coordination and cross-trade clash detection.
- Inflation beyond 30 calendar days from the date of this estimate.
- Supply and installation of wood framing, joists, trusses, or sheathing shown on framing plans.

Change Order Triggers:

- Tonnage increases or decreases exceeding 5% from this estimate.
- Coating system upgrade — for example, from primer-only to hot-dip galvanised.
- Connection type change from simple bolted to AESS or heavily welded.
- Addition of member categories not shown on the current drawing set.
- Revision of drawings that changes profile sizes or eliminates members.

10. FINAL RECOMMENDATION

--- This estimate is preliminary and suitable for budgetary planning only, not for a fixed-price bid. The commercial risk is high due to incomplete drawings. Before this estimate can be considered firm, RFIs must be issued to clarify all missing quantities, lengths, member profiles, and scope conflicts. If tonnage changes by more than 5% or the scope of galvanizing is expanded, a full re-estimate is required.

ACTION SUMMARY

--- Priority Actions for This Analysis

No.	Priority	Action Required	Owner	Deadline	Blocking
1	CRITICAL	Issue RFI to confirm all column lengths and wind post heights.	Estimator	2 days	Firm Quote

No.	Priority	Action Required	Owner	Deadline	Blocking
2	CRITICAL	Issue RFI for a complete intel schedule with marks, sizes, lengths, and locations.	Estimator	2 days	Firm Quote
3	HIGH	Issue RFI to resolve conflicts between plans and schedules (e.g., C8 vs C86).	Estimator	3 days	Detailing
4	HIGH	Issue RFI to clarify coating requirements: Is main steel primed-only and only bearing plates galvanized?	Estimator	3 days	Procurement
5	HIGH	Issue RFI to confirm profile HSS6x6x5/8 is correct or a typo.	Estimator	3 days	Procurement